Lecture 4-1: Mechanics of OLS

Goal: Explain how ordinary least squares fits a line to data: the least-squares principle, the estimators, and the decomposition of variance.

- Mechanics of OLS
  - Principle of least squares
  - Deriving estimators
  - Variance decomposition (ANOVA)
  - R^2 and model fit

Worksheet

1. (*OLS as sample mean*) Show that the sample mean $\bar{Y}$ is the OLS estimator of μ in $Y_i = \mu + u_i$.

2. (*OLS slope as covariance-variance ratio*) Prove that $\hat{\beta}_1 = \dfrac{\widehat{\mathrm{Cov}}(X, Y)}{\widehat{\mathrm{Var}}(X)}$.

3. (*Geometric orthogonality*) Show that the OLS residual vector $\hat{u}$ is orthogonal to the regressor vector x.

4. (*ANOVA decomposition*) Show that $SST = SSE + SSR$, where $SST = \sum(y_i - \bar{y})^2$, $SSE = \sum(\hat{y}_i - \bar{y})^2$, and $SSR = \sum(y_i - \hat{y}_i)^2$.